

CHAPTER # 8:-

Basic Integration
Formula's:

$$(i) \int k dx = kx + C$$

$$(ii) \int \frac{x^n}{f(x)} \frac{dx}{f'(x)} = \frac{x^{n+1}}{n+1} + C$$

$$(iii) \text{Exp: } \int \frac{\sin x}{f(x)} \cdot \frac{\cos x}{f'(x)} dx = \frac{\sin^2 x}{2} + C.$$

$$(iv) \int \frac{\ln x}{x} dx = \int \ln x \cdot \frac{1}{x} \cdot dx = \frac{\ln^2 x}{2} + C.$$

$$(v) \int e^x \cdot dx = e^x + C$$

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$$(vi) \int a^x \cdot dx = \frac{a^x}{\ln a} + C$$

Example:-

$$\int 2^x dx = \frac{2^x}{\ln 2} + C.$$

$$(vii) \int \sin x \, dx = -\cos x + C$$

$$(viii) \int \cos x \, dx = \sin x + C$$

$$(ix) \int \sec^2 x \cdot dx = \tan x + C$$

$$(x) \int \sec x \cdot \tan x \cdot dx = \sec x + C$$

$$(xi) \int \operatorname{cosec} x \cdot \cot x = -\operatorname{cosec} x + C$$

$$(xii) \int \tan x \, dx = \int \frac{-\sin x}{\cos x} \, dx = -\ln \cos x + C.$$

$$= \ln (\cos x)^{-1} + C$$

$$= \ln \frac{1}{\cos x} + C = \ln \sec x + C.$$

$$(xiii) \int \cot x \, dx = \int \frac{\cos x}{\sin x} \, dx = \ln \sin x + C.$$

$$(xiv) \int \sec x \, dx = \ln |\sec x + \tan x| + C.$$

$$(xv) \int \operatorname{cosec} x \, dx = -\ln |\sec x + \cot x| + C.$$

$$(xvi) \int \sinh x \, dx = \cosh x + C$$

$$(xvii) \int \cosh x \, dx = \sinh x + C$$

$$(xviii) \int \frac{dx}{\sqrt{1-x^2}} = \sin^{-1}(x) + C$$

$$(xix) \int \frac{dx}{\sqrt{1-x^2}} = \cos^{-1} x + C$$

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$$(xv) \int \frac{1}{1+x^2} = \tan^{-1}x + C.$$

$$(xvi) \int \frac{dx}{x\sqrt{x^2-a^2}} = \frac{1}{a} \sec^{-1}\left(\frac{x}{a}\right) + C$$

(Proof) Proof:-

$$\int \sinh x \, dx:$$

$$= \int \frac{e^x - e^{-x}}{2} \, dx$$

$$= \frac{e^x + e^{-x}}{2}$$

$$= \cosh x.$$

Example:-

$$\int \frac{dx}{\sqrt{8x-x^2}}$$

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Sol₁

$$= -(8x - x)^2$$

$$= -\{(x^2 - 8x + 16) - (4)^2\}$$

$$= -\{(x - 4)^2 - 16\}$$

$$= +16 - (x - 4)^2$$

$$= + (4)^2 - (x - 4)^2$$

$$\int \frac{dx}{\sqrt{(4)^2 - (x-4)^2}} = \sin^{-1} \left(\frac{x-4}{4} \right) + C \underline{\underline{Ans}}$$

$a^2 - x^2$

$$\int (\cos 2x \cdot \sin x + \sin 2x \cdot \cos x) dx$$

$$\text{Sol}_{12} \int \sin(2x + x) dx$$

$$= \int \sin(3x) dx$$

$$= \frac{-\cos 3x}{3} + C \underline{\underline{Ans}}$$

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Exercise # 8.1

$$(i) \int_0^1 \frac{16x \xrightarrow{f(x)}}{8x^2+2 \xrightarrow{f(x)}} \cdot dx$$

Soln.

$$= \int_0^1 \frac{16x}{8x^2+2} \cdot dx$$

$$= \left| \ln(8x^2+2) \right| \text{ Ans}$$

$$= \ln(8(1)^2+2) - \ln(8(0)^2+2)$$

$$= \ln(10) - \ln(2)$$

$$= \ln \frac{10}{2} = \ln 5 \text{ Ans}$$

$$(iii) \int (\sec x - \tan x)^2 \cdot dx$$

$$\text{Soln: } \int \sec^2 x \cdot dx + \int \tan^2 x \cdot dx - 2 \int \sec x \tan x$$

$$= \tan x + \int (\sec^2 - 1) \cdot dx - 2 \sec x$$

$$= \tan x + \tan x - x - 2 \sec x + C \text{ Ans.}$$

$$(4) \int \frac{1-x}{\sqrt{1-x^2}} \cdot dx$$

Soln-

$$= \int \frac{1}{1-\sqrt{1-x^2}} \cdot dx - \int x(1-x^2)^{-\frac{1}{2}} \cdot dx$$

$$= \int \frac{1}{\sqrt{1-x^2}} + \frac{1}{2} \int \frac{-2x}{\frac{f'(x)}{f(x)}} \cdot dx$$

$$= \sin^{-1} x + \frac{1}{2} \cdot \frac{(1-x^2)^{-\frac{1}{2}+1}}{-\frac{1}{2}+1} + C$$

$$= \sin^{-1} x + \frac{1}{2} \cdot \frac{(1-x^2)^{\frac{1}{2}}}{\frac{1}{2}} + C$$

$$= \sin^{-1} x + (1-x^2)^{\frac{1}{2}} + C \quad \underline{\text{Ans.}}$$

$$5:- \int_{\pi/4}^{\pi/3} \frac{dx}{\cos^2 x \cdot \tan x}$$

Soln-

$$= \int_{\pi/4}^{\pi/3} \frac{1}{\cos^2 x} \cdot \frac{1}{\tan x} dx$$

$$= \int_{\pi/4}^{\pi/3} \frac{\sec^2 x \rightarrow f'(x)}{\tan x \rightarrow f(x)} \cdot dx = \ln |\tan x|^{\pi/3}$$

$$= \ln \left(\frac{\sqrt{3}}{2} \right) - \ln(1)$$

$$(77) \int \frac{e^{-\cot x}}{\sin^2 x} dx$$

Sol-

$$\int e^{-\cot x} \cdot \operatorname{cosec}^2 x dx$$

$$\text{put } -\cot x = t$$

$$\operatorname{cosec}^2 x \cdot dx = dt$$

$$= \int e^{+t} dt$$

$$= e^t + C$$

$$= e^{-\cot x} + C \quad \text{Ans.}$$

Q#9:-

$$\int \frac{dz}{e^z + e^{-z}}$$

$$\text{Soln } \int \frac{dz}{e^z + \frac{1}{e^z}} = \int \frac{e^z dz}{e^{2z} + 1}$$

$$\text{put } e^z = x$$

$$e^z \cdot dz = dx$$

$$= \int \frac{dx}{x^2 + 1}$$

$$= \tan^{-1}(x) + C$$

$$= \tan^{-1}(e^z) + C \quad \text{Ans.}$$

Q# 11:- $\int_{-1}^0 \frac{4 dx}{1 + (2x+1)^2}$

Soln

let $u = 2x+1$

$du = 2x \cdot dx$

$$= 2 \int_{-1}^0 \frac{2 dx}{1 + (2x+1)^2} = 2 \int \frac{du}{1 + u^2}$$

$$= 2 \tan^{-1}(u) + C$$

$$= 2 \tan^{-1}(2x+1) + C$$

Q# 13:- $\int \frac{dt}{1 - \sec t}$

Soln

$$= \int \frac{1 + \sec t}{(1 - \sec t)(1 + \sec t)} \cdot dt$$

$$= \int \frac{1 + \sec t}{(1 - \sec^2 t)} \cdot dt$$

$$= \int \frac{1 + \sec t}{\tan^2 t} dx$$

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$$= - \int \frac{1}{\tan^2 t} dt - \int \frac{\sec t}{\tan t} dt$$

$$= - \int \cot^2 t \cdot dt - \int \frac{\overset{1}{\cancel{\cos t}}}{\sin t / \cancel{\cos t}} \cdot dt$$

$$= - \int \cot^2 t \cdot dt - \int \operatorname{cosec} t \cdot dt$$

$$= - \int (\operatorname{cosec}^2 t - 1) dt - \int \operatorname{cosec} t \cdot dt \underline{\underline{Ans}}$$